FINAL BIOINFORMATICS REPORT

Basic Sequence Analysis and Visualization of Human Insulin (INS) mRNA

Internship: Swynex Bioinformatics Internship
Task: Task 4 — Final Bioinformatics Report
Organism: Homo sapiens
Gene: INS (Insulin)
Transcript accession: NM_000207.3
Database: National Center for Biotechnology Information (NCBI)

1. Abstract

Bioinformatics combines biology, computer science, and data analysis to study biological information. This project involved selecting a human insulin mRNA sequence from the NCBI database, retrieving the sequence in FASTA format, performing basic sequence analysis using Python, and creating a bar chart using Matplotlib. The sequence analysis included calculating nucleotide counts, sequence length, and GC content. A separate illustrative gene-expression visualization was created to practise graphical representation and interpretation. This project demonstrates the use of publicly available biological data and basic computational tools in bioinformatics.

2. Research Question

What are the basic sequence characteristics of human insulin (INS) mRNA, and how can Python be used to analyse biological sequence data and visualize illustrative values?

3. Introduction

Bioinformatics is an interdisciplinary field that uses computational methods to collect, process, analyse, and interpret biological data. It is widely used in genomics, molecular biology, pharmaceutical research, and biotechnology.

The insulin gene (INS) is associated with the production of insulin, a hormone that helps regulate blood glucose levels. Studying its nucleotide sequence provides an opportunity to understand how biological sequence information can be retrieved and analysed using computational tools.

In this project, the human insulin mRNA sequence was selected from NCBI. Python and the Biopython library were used for basic sequence retrieval and analysis. Matplotlib was used to create a bar chart illustrating gene-expression values.

4. Objectives

The main objectives were:

1. To identify and retrieve a biological dataset from NCBI.
2. To obtain the human insulin mRNA sequence in FASTA format.

3. To use Python and Biopython for basic sequence analysis.
4. To calculate sequence length, nucleotide counts, and GC content.
5. To create a bar chart using Python's Matplotlib library.
6. To interpret the results and identify the limitations of the analysis.

5. Materials and Tools

The following resources and tools were used:

- NCBI: For retrieving the biological sequence.
- Python: For computational analysis.
- Google Colab: For executing Python code online.
- Biopython: For handling biological sequence data.
- Matplotlib: For generating a bar chart.
- Google Docs and Google Drive: For preparing and organizing the report and submission materials.

6. Dataset Selection

The dataset was obtained from the National Center for Biotechnology Information (NCBI) database.

The selected record was the human insulin (INS) transcript, variant 1, with accession number NM_000207.3.

Dataset details:

- Organism: Homo sapiens
- Gene: INS
- Molecule: mRNA
- Transcript: Variant 1
- Accession number: NM_000207.3
- Sequence length: 465 nucleotides
- Format: FASTA

The FASTA format contains a header identifying the sequence and a sequence section containing nucleotide characters.

This dataset was selected because it represents a biologically relevant human gene and provides a suitable example for learning basic biological sequence analysis.

7. Methodology

The project was completed in the following stages.

7.1 Biological Dataset Retrieval

The NCBI database was accessed to locate the human insulin mRNA record. The transcript accession NM_000207.3 was identified, and the sequence was retrieved in FASTA format.

7.2 Sequence Processing

Python and the Biopython library were used to retrieve and process the nucleotide sequence.

The sequence was examined to determine its length and nucleotide composition. The nucleotide counts were calculated for adenine (A), thymine (T), guanine (G), and cytosine (C).

7.3 GC Content Calculation

GC content represents the percentage of guanine and cytosine nucleotides in a sequence.

It was calculated using the formula:

GC content (%) = [(Number of G + Number of C) / Total sequence length] × 100

7.4 Data Visualization

A bar chart was created using Matplotlib in Python.

Five illustrative samples were assigned values of 12, 18, 15, 25, and 20. The chart was labelled with a title, sample names, and an expression-level axis.

The visualization was used to practise graphical presentation and interpretation. The values were illustrative and were not obtained from actual gene-expression experiments.

8. Results

8.1 Sequence Analysis Results

The basic sequence analysis produced the following results:

Parameter| Result
Organism| Homo sapiens
Gene| INS
Transcript accession| NM_000207.3
Sequence length| 465 nucleotides
Adenine (A) count| 91
Thymine (T) count| 77
Guanine (G) count| 141
Cytosine (C) count| 156
GC content| 63.87%

The nucleotide counts add up to 465, matching the reported sequence length.

GC content was calculated as follows:

GC content = [(141 + 156) / 465] × 100

GC content = (297 / 465) × 100

GC content = 63.87%

These results describe the basic nucleotide composition of the selected sequence.

8.2 Visualization Results

The illustrative gene-expression bar chart contained the following values:

Sample	Illustrative value
Sample 1	12
Sample 2	18
Sample 3	15
Sample 4	25
Sample 5	20

Sample 4 had the highest illustrative value of 25, while Sample 1 had the lowest value of 12.

The bar chart demonstrated how Matplotlib can be used to compare values visually and identify the highest and lowest observations.

Important distinction: These sample values were manually supplied for visualization practice. They were not measured expression levels of the INS gene and cannot be used to draw biological conclusions about insulin gene expression.

8.3 Overall Findings

The project demonstrated the retrieval and basic computational analysis of a biological sequence. The sequence length, nucleotide counts, and GC content were determined using Python-based methods.

The visualization exercise demonstrated how numerical values can be presented in a bar chart and interpreted. Together, these activities provided practical experience with biological databases, computational sequence handling, and scientific visualization.

9. Limitations

The project had the following limitations:

1. The sequence analysis was limited to basic nucleotide composition and sequence length.
2. The analysis did not investigate gene regulation, protein structure, or biological activity.

3. The illustrative visualization used manually assigned values rather than experimental gene-expression measurements.
4. The chart cannot establish statistical significance, biological causation, or differences in actual INS gene expression.
5. The project did not include experimental validation in a laboratory.
6. The conclusions are limited to the analyses performed and should not be interpreted as evidence of clinical or therapeutic effects.

10. Applications in Biotechnology and Pharmaceutical Research

Basic bioinformatics techniques are useful in biotechnology and pharmaceutical research because they help researchers retrieve and organize biological information.

Sequence analysis can support preliminary investigations of genes and transcripts. Nucleotide composition calculations provide basic descriptive information, while visualization tools help communicate numerical results clearly.

More advanced studies may use sequence alignment, gene annotation, protein analysis, and experimental expression data to investigate biological questions. These methods can contribute to research in molecular biology, disease mechanisms, and drug discovery.

However, the present project is an introductory computational exercise and does not demonstrate a drug discovery outcome.

11. Conclusion

This project provided practical experience in retrieving and analysing a biological dataset using publicly available resources and computational tools.

The human insulin mRNA transcript, accession NM_000207.3, was selected from NCBI. Its reported length was 465 nucleotides, and its calculated GC content was 63.87%. Python and Biopython were used for sequence-related tasks, while Matplotlib was used to create an illustrative bar chart.

The project also highlighted the importance of distinguishing real biological measurements from example data. Overall, it strengthened my understanding of biological sequence retrieval, basic computational analysis, data visualization, and the limitations of interpreting bioinformatics results.

12. References

1. National Center for Biotechnology Information (NCBI). Nucleotide database. Homo sapiens insulin (INS) transcript variant 1, accession NM_000207.3.
2. Cock, P. J. A., et al. (2009). Biopython: Freely available Python tools for computational molecular biology and bioinformatics. Bioinformatics, 25(11), 1422–1423.
3. Hunter, J. D. (2007). Matplotlib: A 2D graphics environment. Computing in Science & Engineering, 9(3), 90–95.
4. Python Software Foundation. Python documentation.

5. Matplotlib Development Team. Matplotlib documentation.

13. Declaration

This report summarizes the biological dataset retrieval, basic sequence analysis, and illustrative visualization activities completed as part of the bioinformatics internship tasks. The results and limitations are presented according to the work performed.

End of Report